Academic Book Recommender System for University Students Using Content-Based Filtering

Graduation Project – Bachelor of Computer Science

Department of Computer Science

Majmaah University



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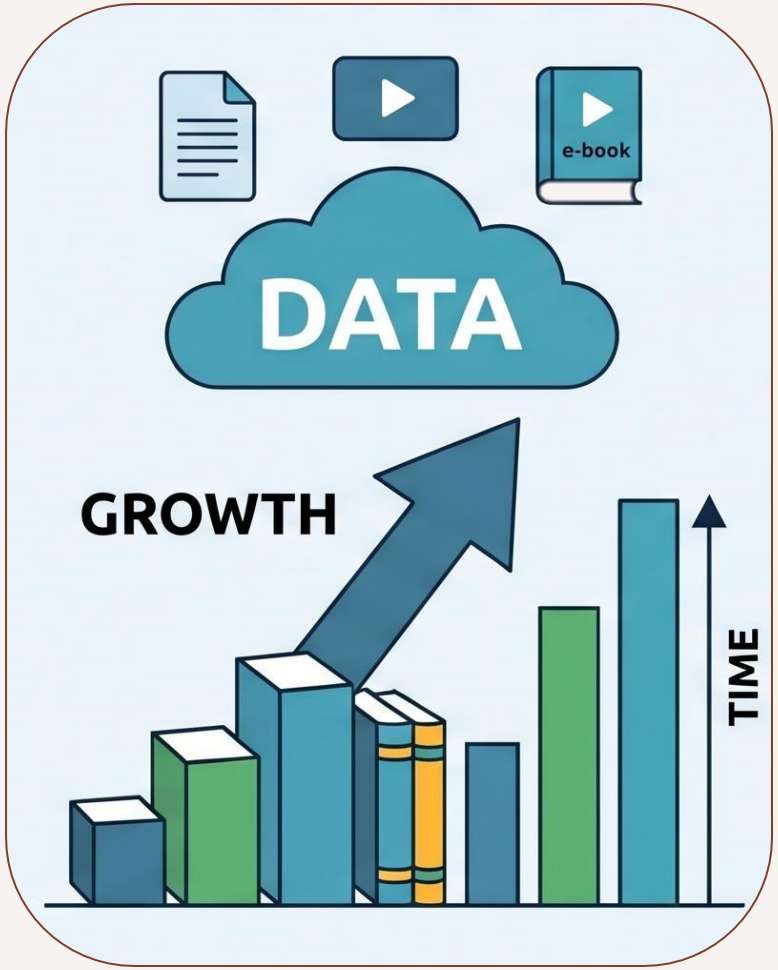
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.1Introduction & Background

Project Background

- Proliferation of digital libraries increases resource search friction.
- Traditional university catalogs offer rigid, non-personalized exploration paths.
- Al-Warraq aims to automate scholarly discovery by matching specific student profiles directly with highly relevant domain materials.



2. Introduction & Motivation

The Academic Shift: Massive expansion of scholarly digital libraries increases resource search friction.

The Goal: Automate resource discovery within narrow computing domains (e.g., Computer Science).

Al-Warraq Solution: An intelligent web platform that matches individual student profiles and specific disciplines directly with relevant book metadata.

3. Problem Statement

Information Overload: The massive explosion of digital book data makes it extremely difficult for students to find relevant academic resources.

Rigid Search Systems: Traditional libraries rely on basic keyword searches, ignoring semantic context and lacking personalization.

Wasted Time & Effort: Students spend excessive hours manually filtering through databases, which negatively impacts their academic productivity.

The Solution

Developing an intelligent, content-based recommendation system to provide personalized book suggestions and eliminate manual search friction.

3. Project Scope & Boundaries

In-Scope Features:

- Core NLP-driven recommendation engine.
- Interactive AI Conversational Librarian.
- User Profile creation & Academic interest tuning.
- Persistent database tier for saved favorites.

Out-of-Scope Elements:

- Standalone Mobile Apps (System is purely a web prototype).
- Advanced Hybrid filtering (Targeted for initial deployment stability).
- Cross-Cultural Recommendations in Multiple Languages

Expected Outcomes

1. **Experimental Platform:** Fully integrated Python/Flask web app.
2. **Data Sufficiency:** Ingests 5K+ records showing complete metadata (Title, Author, Category).
3. **Course Alignment:** Directly addresses student specialized disciplines.

4. System Requirements (FRs & NFRs)

Functional Requirements (FR):

1. User profile interest tagging and customization.
2. TF-IDF vector extraction and Cosine Similarity calculation.
3. Persistent storage for individual user favorites list.
4. Automated Chatbot dialogue answering queries based on book descriptions or knowledge.

Non-Functional Requirements (NFR):

Performance: Search operations must resolve in under 150ms.

Scalability: System must handle up to 15 concurrent users stably.

Availability: Target of 95% total server availability.

Safety: Strictly deterministic AI responses (0.1 Temperature).

5. Base Scope vs Advanced Improvements

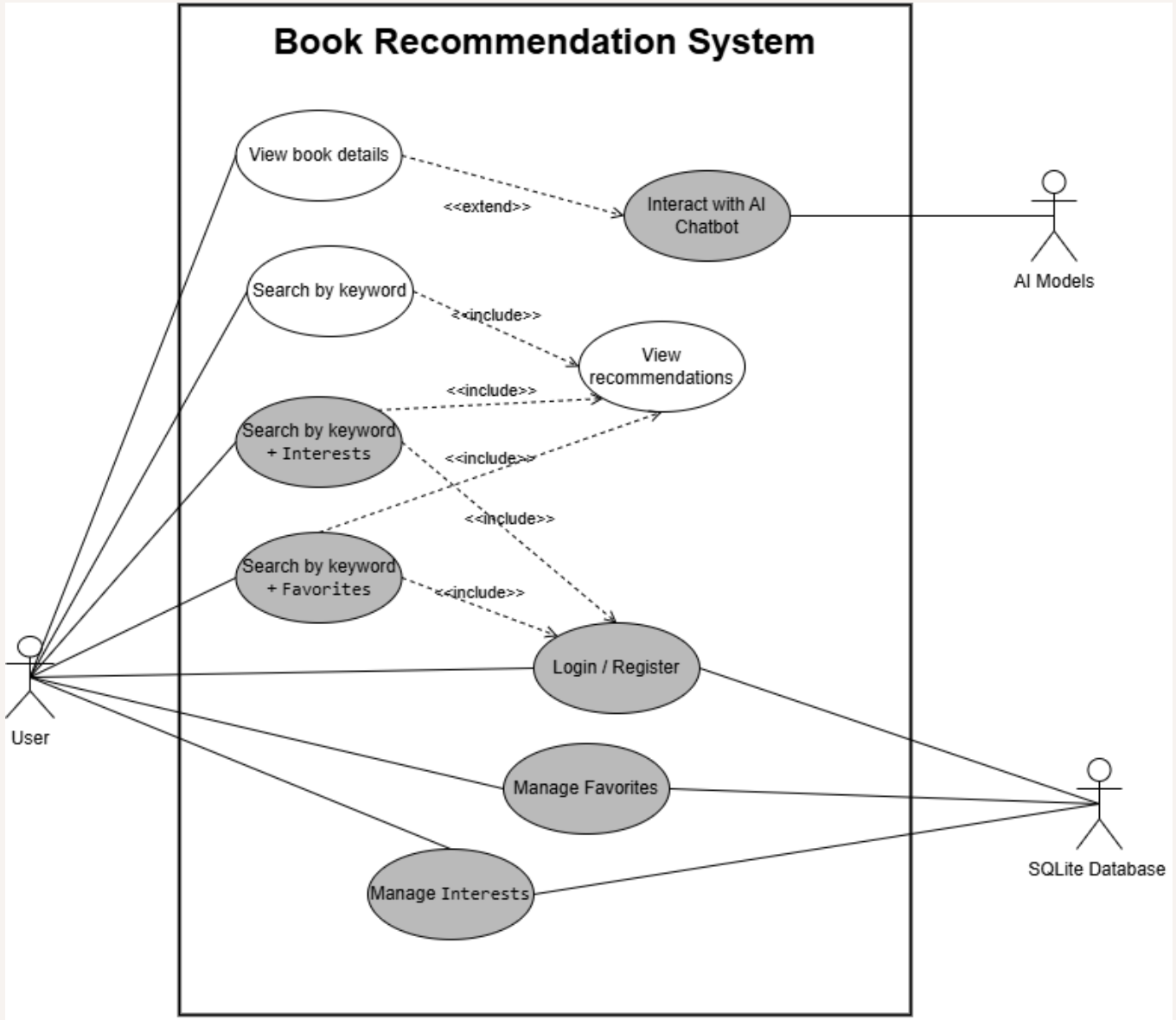
Standard Expected Scope:

- Basic keyword mapping and lookup.
- Simple static data storage and retrieval.

Al-Warraq Engineering Enhancements (Beyond the Scope):

- **Multi-Phase Scoring:** Combining raw queries with profile interests and saved favorites weights.
- **NLP Query Resilience:** Real-time typo correction and acronym expansion.
- **In-Memory Optimization:** Loading pre-computed matrix configurations entirely in RAM .
- **High-Availability AI Infrastructure:** Automated multi-cloud fallback routing.

6. System Architecture & Tech Stack



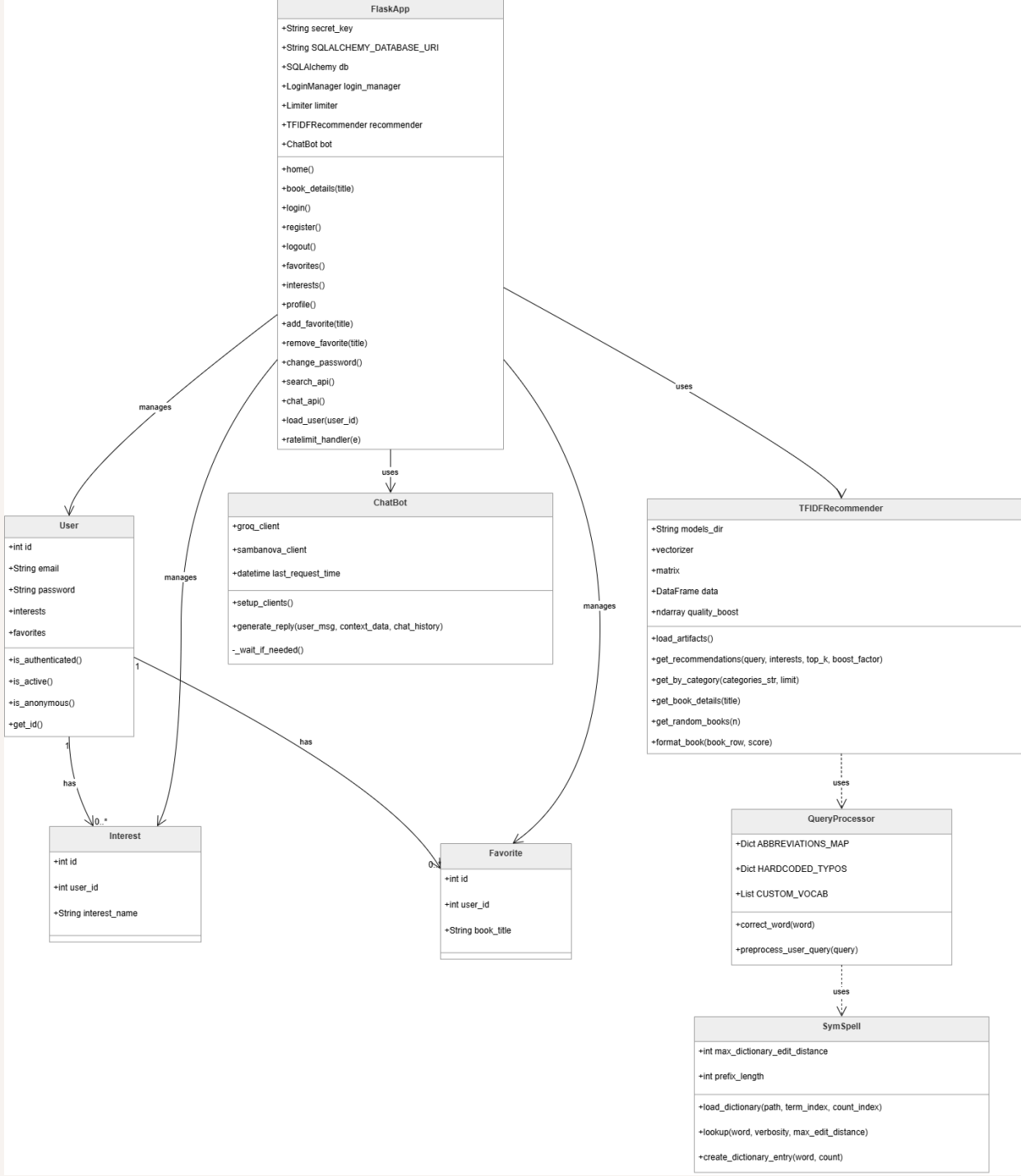
This diagram highlights the system's key enhancements, specifically the personalized search features and the AI chatbot integration.

6. System Architecture & Tech Stack

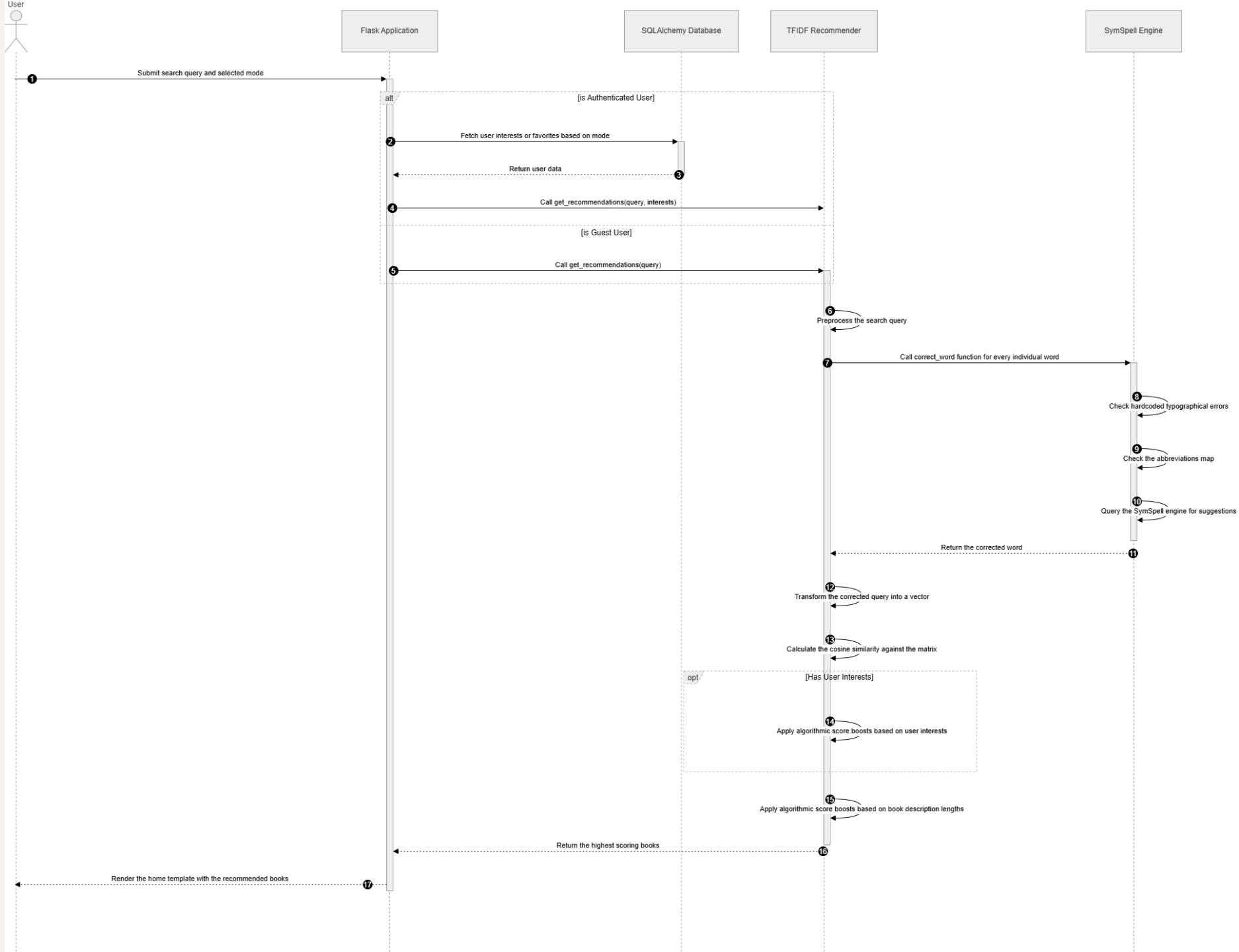
System Architecture

•Infrastructure:

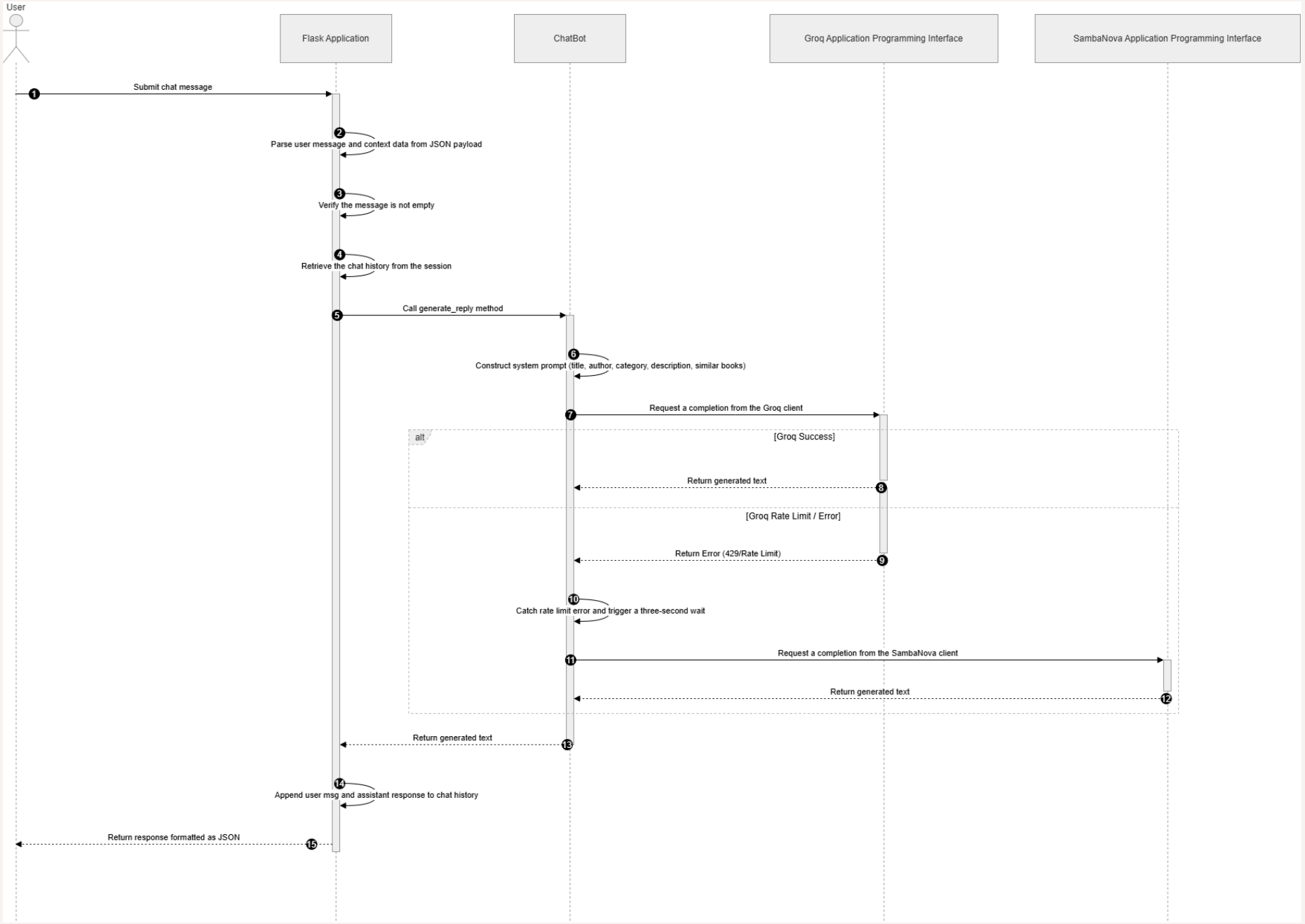
- **Backend:** Python / Flask Framework
- **Database:** SQLite with SQLAlchemy
- **NLP Engine:** Scikit-learn (Vectorization)
- **AI Models:** Llama 3.3 via Dual API Cloud Nodes



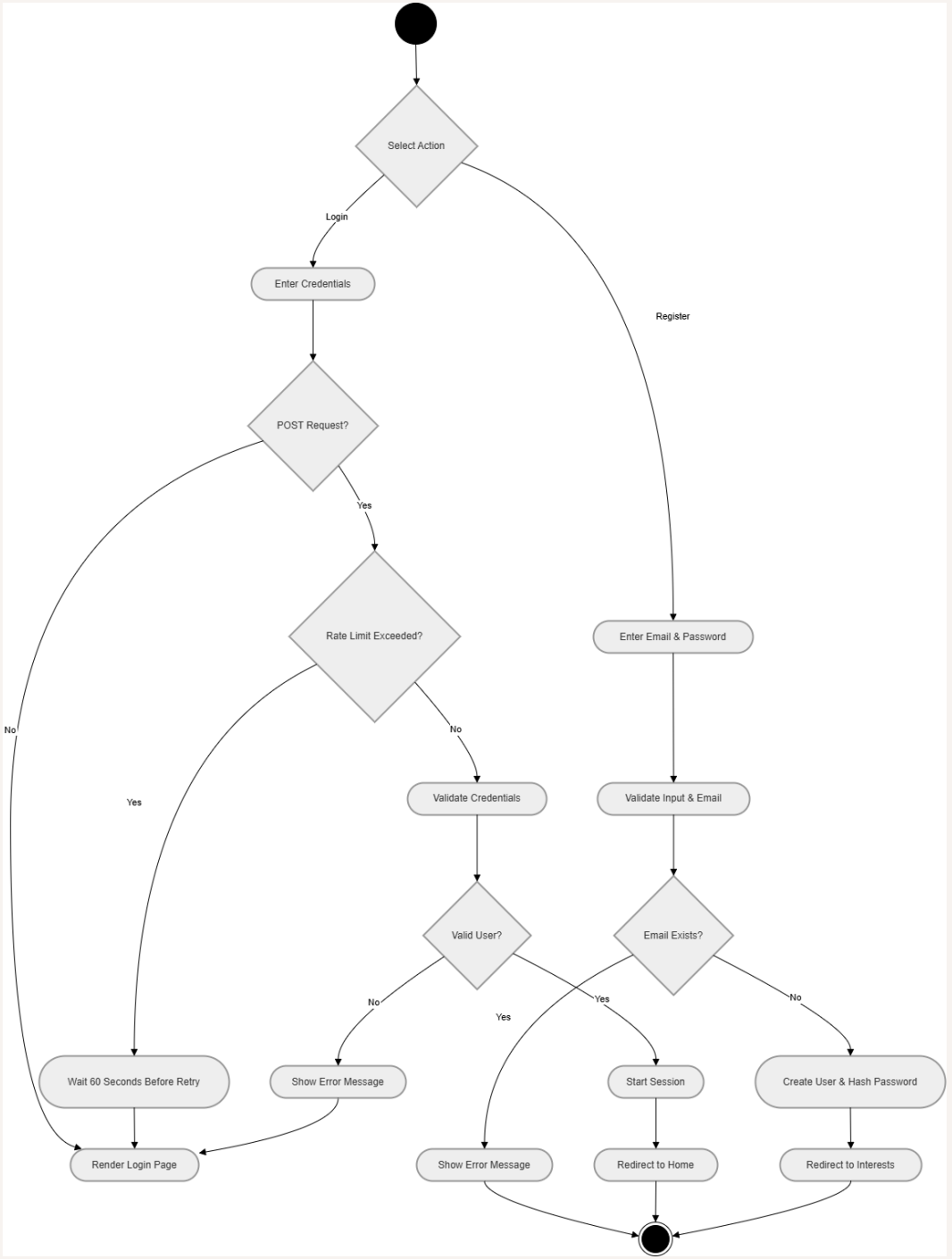
٧. Sequence Diagram



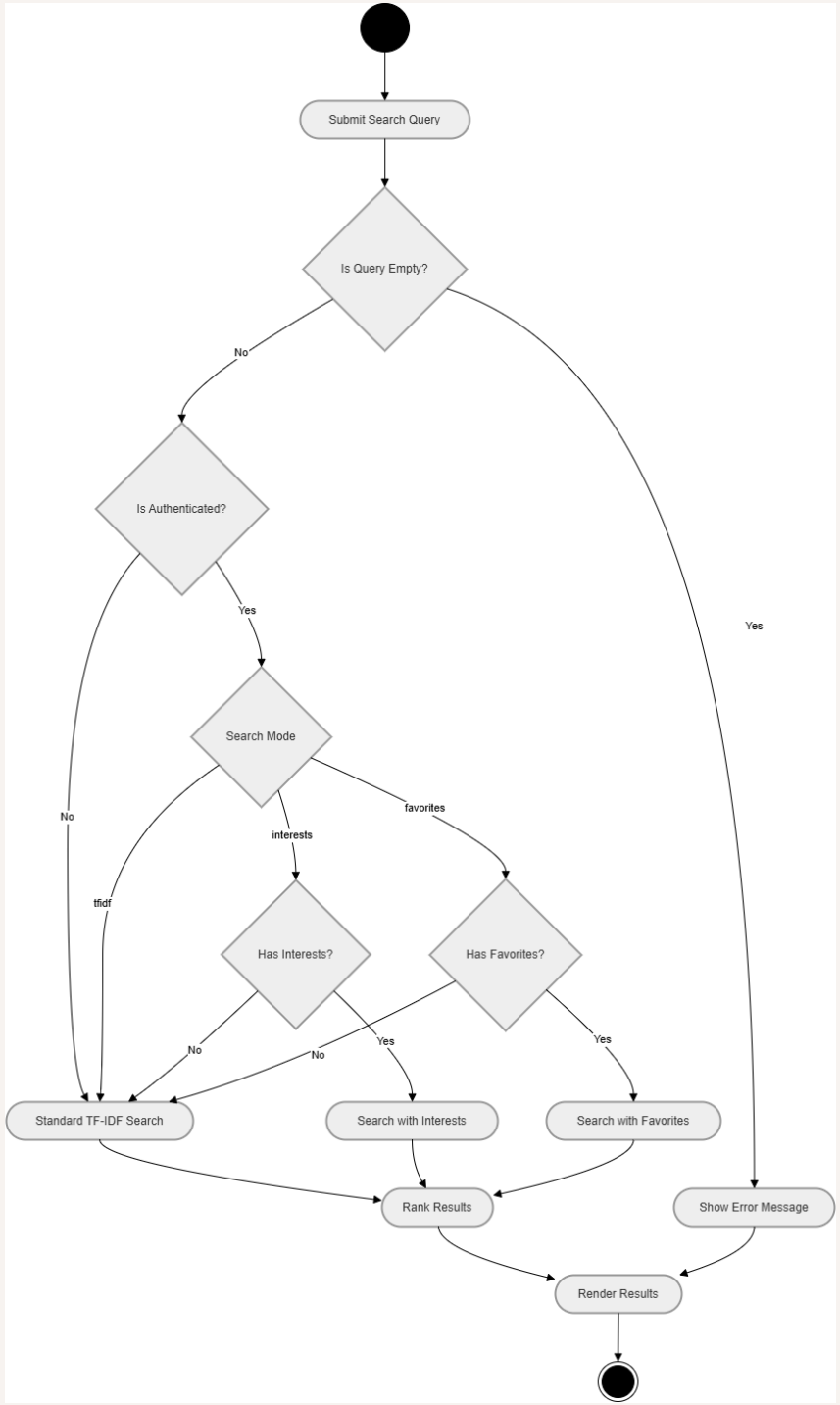
٧. Sequence Diagram



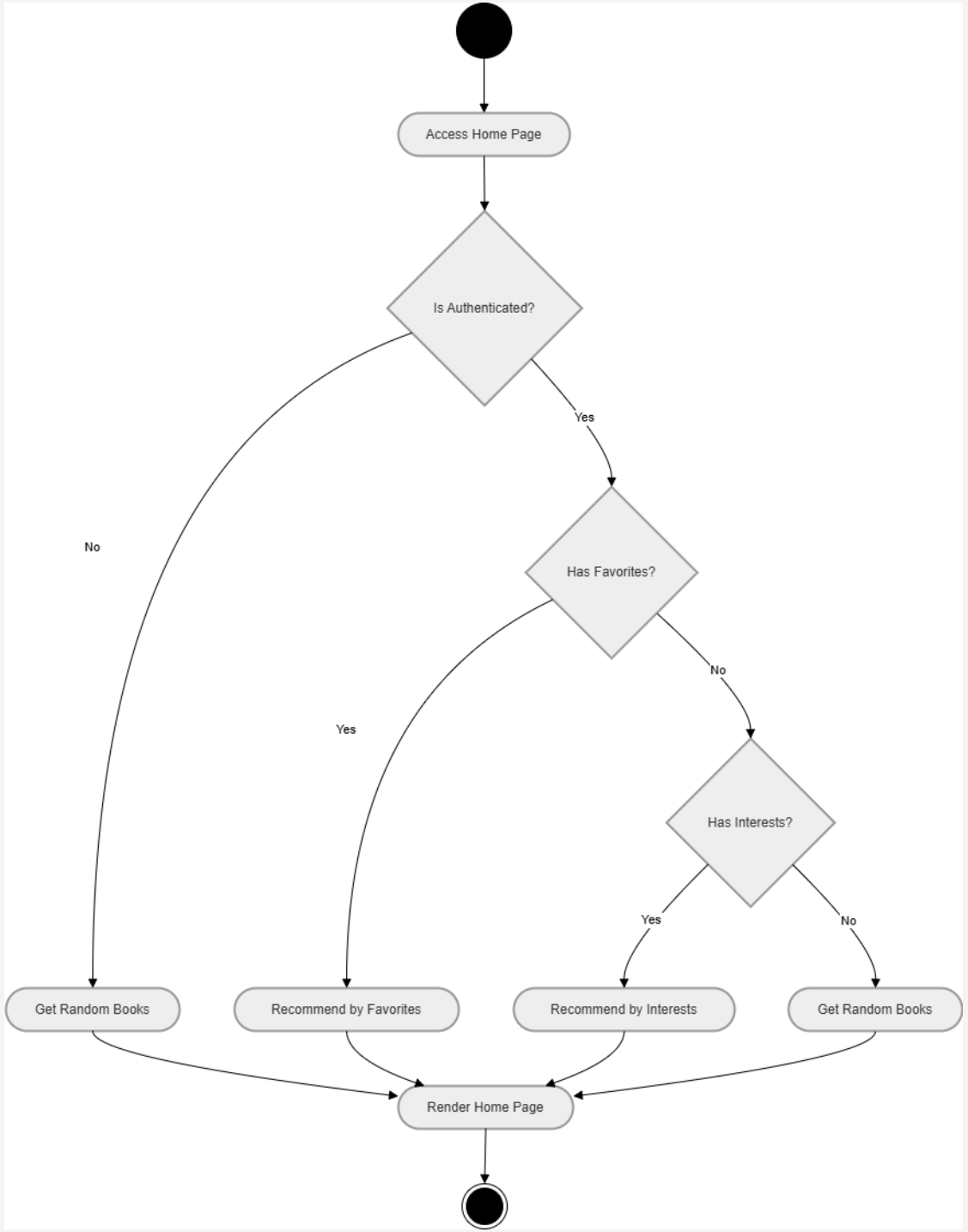
٨. Process Logic: Auth & Search Pipelines



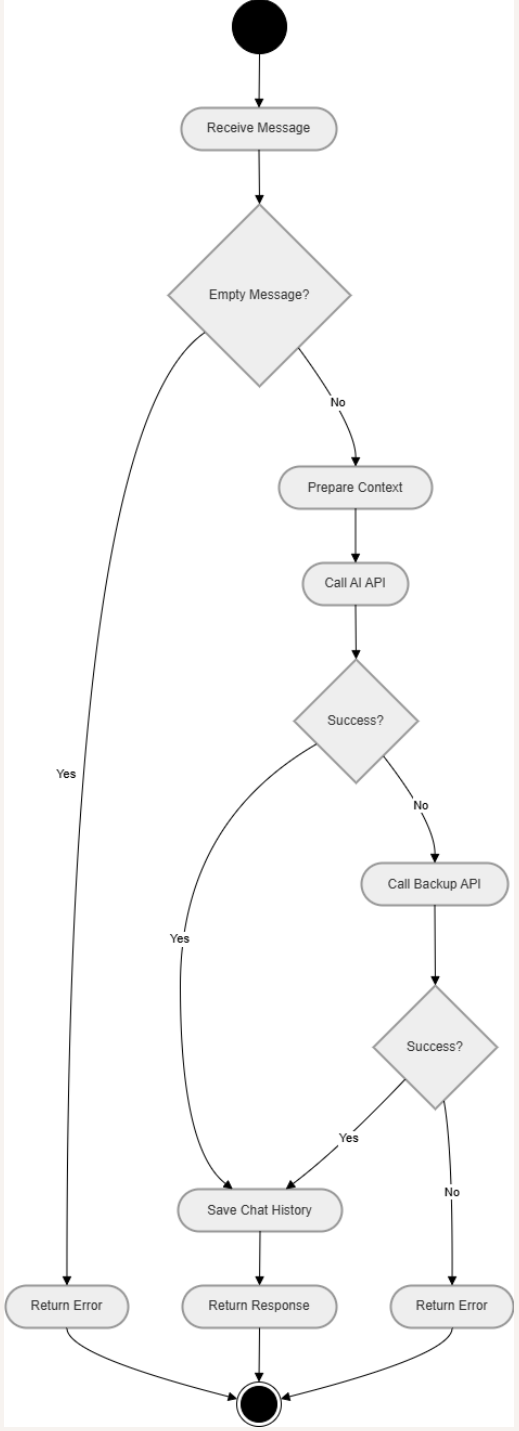
٨. Process Logic: Auth & Search Pipelines



ๆ. Process Logic: Recommendation



ๆ. Process Logic: Chatbot



١٠. Data Collection & Preprocessing

1

- **Data Collection:** The initial dataset was extracted from the Open Library API. Because of the high volume of requests across more than ٨٠ academic sub-disciplines, concurrent Python scripts were used to pull the data efficiently .

2

- **Data Cleaning:** The raw data was converted into structured tables using pandas drowyek ‹cimedaca yltcirts deniamer mrotfalp eht erusne oT . ekil sdrow gniniatnoc tnetnoc cimedaca-non evomer ot deilppa saw gniretlfi ."yretsym" ro "‹noticfi" "‹levon"

3

Data Quality Improvement (Handling Missing Data):
Combined the book's title, subjects, and tags to fill in missing descriptions.
Used Google Books API, which increased available descriptions by .٢٠%

١١. Core System Implementation: Two-Phase Architecture

- **Phase ١: Ofline training:**

Pre-computing TF-IDF matrices for١٠,٠٠٠ vocabulary tokens based on book metadata (Titles, Authors, Descriptions, Categories, Tags).

- **Phase 2: Real-Time Inference**

Executing Cosine Similarity instantly in RAM using pre-loaded weights, completely avoiding database delays.

$$w_{t,d} = TF_{t,d} \times \log \frac{N}{DF_t}$$

$$\text{Similarity}(A, B) = \frac{A \cdot B}{||A||\ ||B||} = \frac{\sum_{i=1}^{n} A_i . B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}$$

١٢. Improvement: NLP Query Resilience

SymSpell Error Correction:
Rectifies typographical errors in under 0.01ms.
Example: Maps "Artficial" ➔ "Artificial" before vector processing.

Acronym Query Expansion:
Maps technical shortcuts (e.g., 'AI', 'NLP') to full linguistic expressions. Drastically increases token match coverage inside book metadata.

١٢. Improvement: Mathematical Model - Personalized Boosting



Dynamically adjusting the base Cosine Similarity score to inject user profile context without relying on collaborative data.

$$Boost_{Category} = \beta \times 0.3 \quad (\textit{if category matches})$$

$$Boost_{Tag} = \beta \times 0.5 \quad (\textit{if tag matches via Regex})$$

$$Score_{Personalized} = Score_{Base} + Boost_{Category} + Boost_{Tag}$$

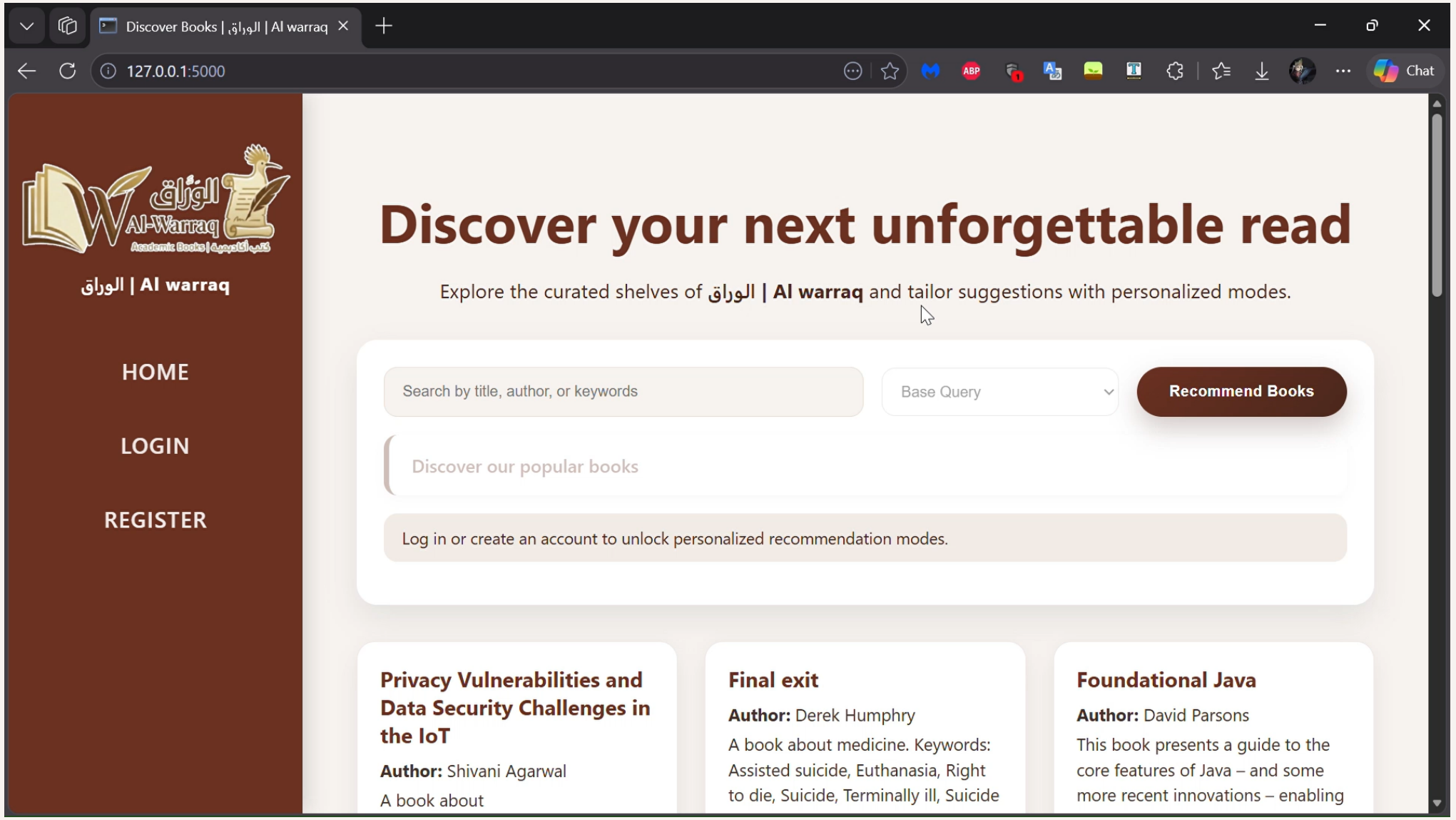
١٢. Improvement: Multi-Cloud AI Chatbot



Dual-Node Framework:
Primary: Groq Cloud API (Ultra-low latency).
Fallback: SambaNova API (Instantly takes over during network drops or quota limits).

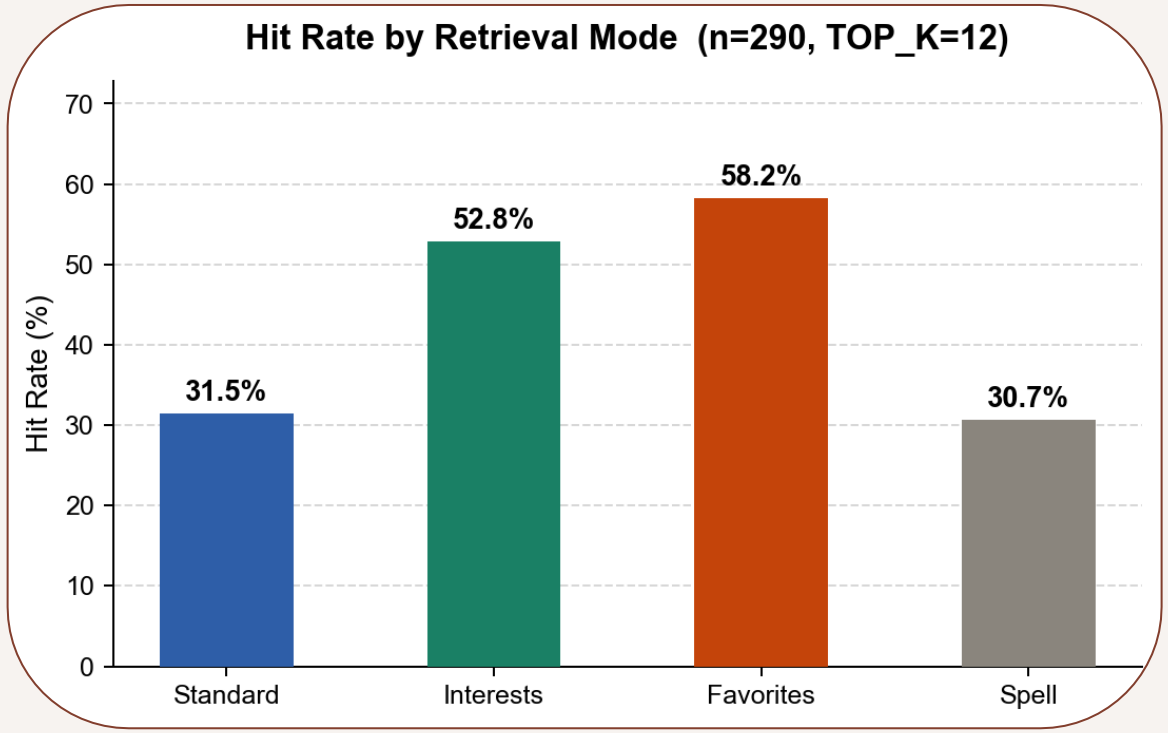
Factual Grounding:
Injects exact database book metadata into the system prompt. reduces the risk of hallucinations and adversarial prompt injections.

١٣. Demo video



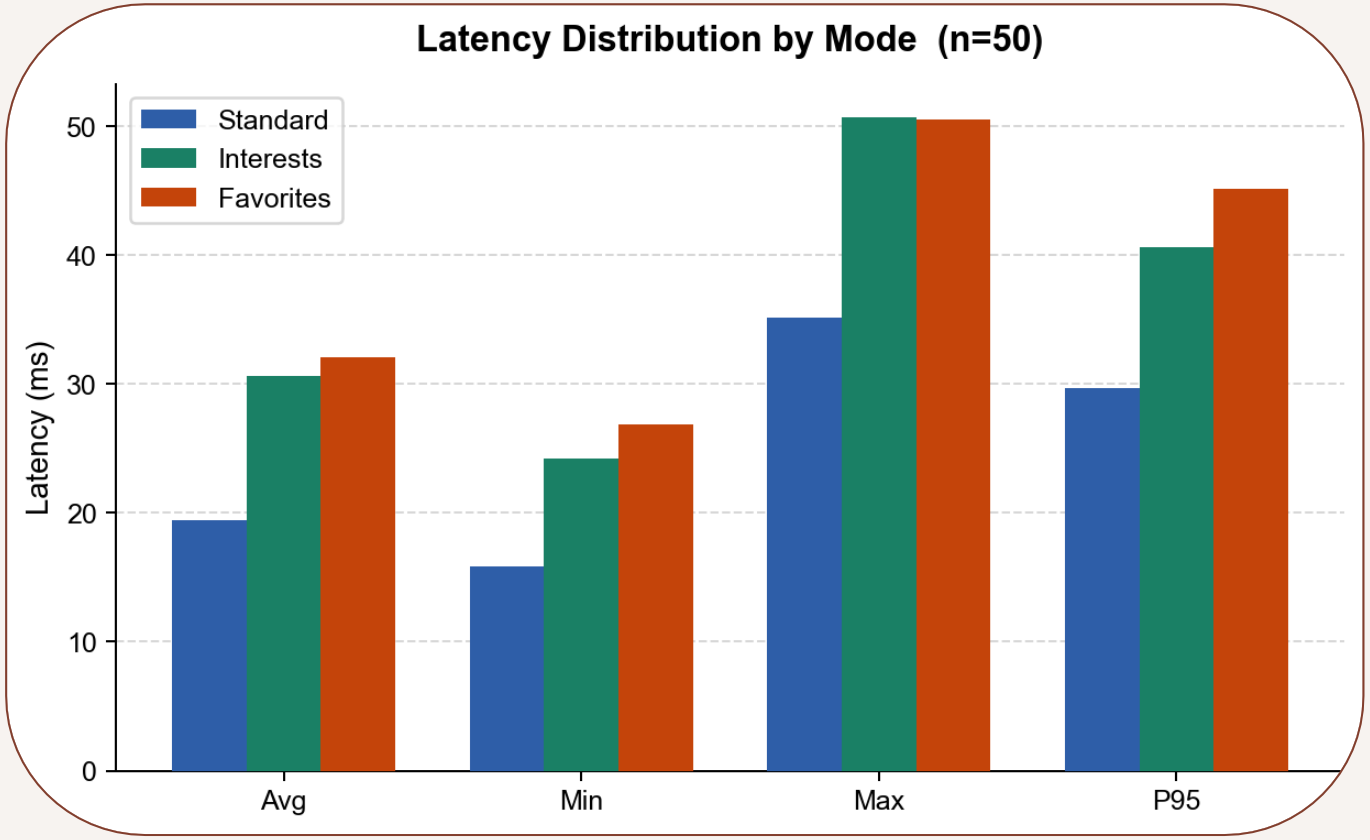
14. Evaluation: Personalization Accuracy Lift

- **Empirical Validation (290 Queries):**
- **Standard Mode:** 31.5% Hit Rate.
- **Interests-Based Boost:** 52.8% Hit Rate (+21.3% lift).
- **Favorites-Based Re-ranking:** 58.2% Hit Rate (+26.8% lift).



14. Evaluation: Personalization Accuracy Lift

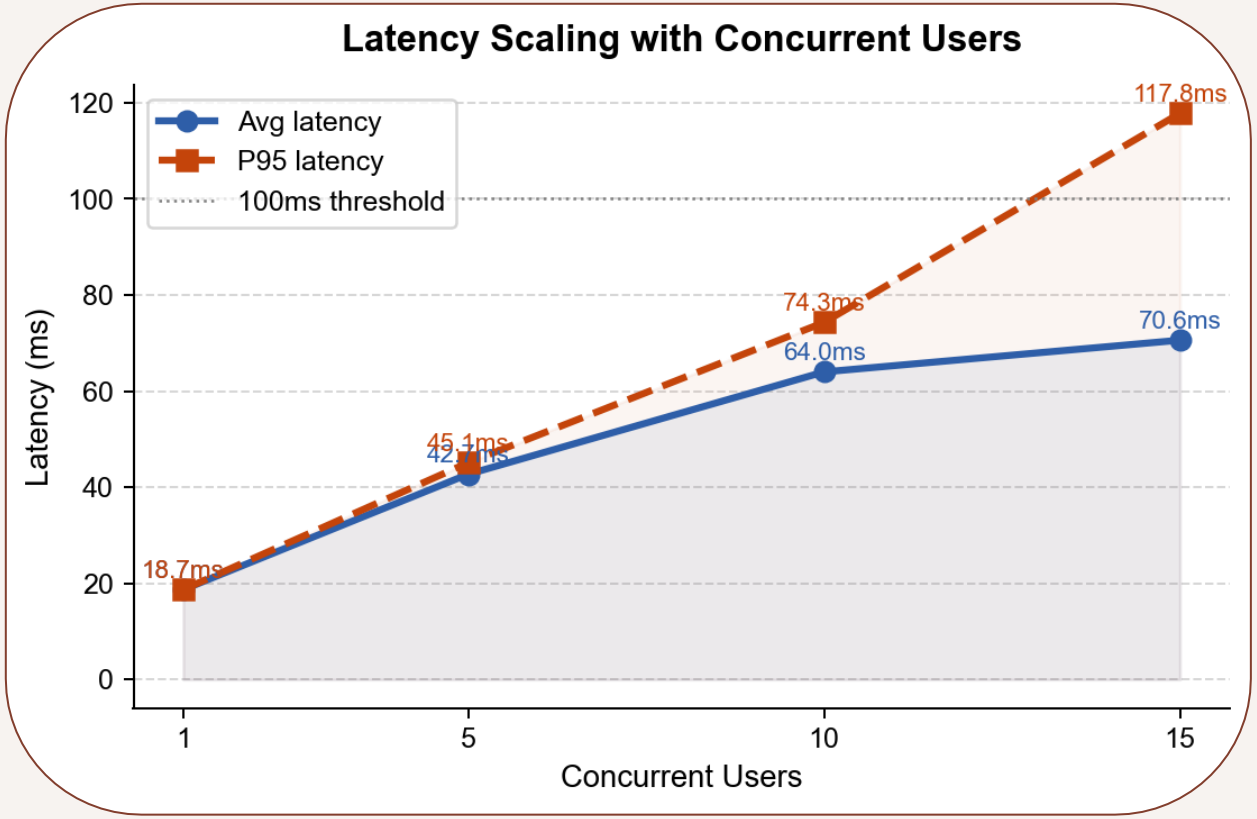
- The evaluation utilized a TOP_K parameter of 12 and measured Accuracy (Hit Rate, NDCG, Precision@1) and Speed (Average Latency).
- **Conclusion:** Persistent profile metrics dramatically increase recommendation relevance.



14. Evaluation: Concurrency & Latency

Stress Test Benchmarks:

- **1 Active User:** 18.7ms average response time.
- **15 Concurrent Users (Peak Load):** 70.6ms response time.
- **Conclusion:** Latency scales safely and efficiently well below the strict 150ms NFR boundary.



1º. Conclusion & Future Roadmap

Hybrid Models: Integrate User-to-User Collaborative Filtering layers.

Deep RAG Upgrade: Transition the Chatbot to analyze specific book chapters (Retrieval-Augmented Generation).

Campus Sync: Direct database linkage with the physical University Library inventory.

Thank you for your attention

We hope this project reflects the skills and knowledge we developed throughout our academic journey.